pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Start Here: Python, Notebooks, and Libraries

WEEK 1 · DAY 0

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

What you will know by the end

What Python is

A language for instructions, and the engine that runs them.

```
print("Good morning, bond desk")
```

What a notebook is

A column of cells: markdown for text, code for Python.

```
Shift + Enter
```

What a .py file is

Plain text that holds only Python, run top to bottom.

```
python half_coupon.py
```

What a library is

Code someone already wrote: install once, import each time.

```
import pandas as pd
```

What Python is

You write

instructions in Python

```
print("Good morning, bond desk")
```

Python runs them

the engine on your computer

```
reads each line, carries it out
```

A result comes back

under the cell

```
Good morning, bond desk
```

Excel is the grid and the engine in one application.

Python separates them: Python is the engine, and in this course you write in a notebook.

What a notebook is

Part	What it is
Cell	One box in the column: a code cell or a markdown cell
Code cell	Python instructions; running it hands them to the kernel
Markdown cell	Text: headings, explanations, tables. Python ignores it
Output	What a code cell shows under itself after it runs
Kernel	The copy of the Python engine behind the notebook
.ipynb file	The saved notebook: cells, code, and outputs together

Opened in VS Code or Colab. Like a workbook where notes and formulas sit on one page, in order.

Two kinds of cells

Markdown cell: text

```
## Morning check
**coupon** in bold
* a bullet point
```

Type the symbols; run the cell and they become formatting. Double-click to edit.

Code cell: Python

```
coupon = 5.25
coupon / 2
# 2.625 appears underneath
```

Run it, and the result appears underneath, with [1], [2] showing the order cells ran.

Run one cell	Run every cell
Shift + Enter, or the play button beside the cell	Run All: VS Code toolbar; Colab Runtime menu (Ctrl + F9)

A notebook and a .py file

	Notebook (.ipynb)	Python file (.py)
Holds	text, code, and results	code only
Runs	one cell at a time, when you choose	the whole file, top to bottom
Good for	working step by step, showing results	code you keep and use again
Opened in	VS Code, Colab	any editor; run by Python

```
coupon = 5.25
print("Half the coupon:", coupon / 2)
```

half_coupon.py: two lines of Python and nothing else. Run, it prints Half the coupon: 2.625.

Libraries: code someone already wrote

1. Install, once

Copies the library onto your computer. The standard library comes with Python; others come from PyPI with pip.

```
%pip install --quiet pandas
```

2. Import, each time

Makes it available in this notebook. The kernel starts with a clean memory, so import runs every time.

```
import pandas as pd
```

Built in	Standard library	Installed with pip
print, len, sum, max, min	statistics, datetime, math	pandas, numpy, matplotlib

```
Missing or misspelt: ModuleNotFoundError: No module named 'bondtools'
```

Excel side by side

In Excel	In Python
A workbook file (.xlsx)	A notebook file (.ipynb)
Excel, the application that calculates	Python, the engine that runs the code
A cell holding =5.25/2	A code cell holding 5.25 / 2
A text box or a note on the sheet	A markdown cell
F9 to calculate	Run All
SUM, AVERAGE	sum, statistics.mean
An add-in, such as the Analysis ToolPak	A library, installed and then imported
A VBA module	A .py file

An Excel formula recalculates by itself. A Python cell runs only when you run it.

Where the name comes from

Guido van Rossum, CWI

Started during the 1989 Christmas holidays at CWI in the Netherlands. Posted to USENET in February 1991, after just over a year of work.

Not the snake

He was reading the scripts of Monty Python's Flying Circus, the BBC comedy series from the 1970s, and wanted a name that was short, unique, and slightly mysterious.

Sources: docs.python.org/3/faq/general.html and docs.python.org/3/license.html, read 25 October 2026.

The AI check

Every notebook ends with code written in the style of an AI assistant. It has one bug. Find it.

1

Read

Read the code before you run it.

2

Predict

Decide what it should produce.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out yourself.

Code that runs is not the same as code that is right.

What you can do now

- Say what Python is: a language, and the engine that runs it
- Tell a markdown cell from a code cell
- Run a cell with Shift + Enter, and a notebook with Run All
- Say what a .py file is, and when to use one
- Import a library, and read the error when one is missing

Day 1 Introduction to Python: variables, lists, loops, and functions, on four bonds